

YURII OKSAMYTNYI

AI/ML Systems Engineer | Enterprise AI & Agentic Systems

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SUMMARY

AI/ML systems engineer currently consulting with **HSBC** on AI-component discovery, source-data quality and governed AI assistance. Designs and delivers enterprise AI platforms, agentic workflows, retrieval systems and multilingual voice applications across financial services and software products. Combines architecture, Python back-end development, evaluation and full-stack delivery with a focus on traceability and reliable user workflows.

EXPERIENCE

AI Engineering Consultant — HSBC engagement

Independent Consulting

Nov 2025 – Present

London, UK / Remote

Helping internal teams discover reusable AI capabilities and improve the quality of the information used to recommend them.

- **Built and demonstrated a searchable AI-component catalogue** with versioned PostgreSQL storage, source provenance, automated quality checks, FastAPI endpoints and React interfaces. In the evaluated demo snapshot, approximately **33%** of imported records were excluded from recommendation eligibility by quality checks.
- **Corrected silent relationship-data loss** in a knowledge-ingestion pipeline, recovering **100% of known affected values** in the validated corpus and defining stronger source-validation and controlled-refresh requirements.
- **Designed conversational access to internal documentation and metadata** using Google ADK and MCP orchestration, persistent sessions, semantic caching and multi-provider gateway integration.
- **Prototyped evidence-grounded recommendations** for reusable agents, skills and MCP capabilities; designed governed metadata enrichment that combines deterministic checks, catalogue-constrained model suggestions and human approval.

Lead AI Engineer & Systems Designer (Consultant)

Independent

Apr 2022 – Present

London, UK (Hybrid)

Selected additional consulting across enterprise software, legal technology and data products. Work spans **agentic workflows**, **retrieval-augmented generation (RAG)**, **real-time voice AI** and the cloud infrastructure required to support them.

- **Architected and delivered multilingual legal-intake voice AI for Jupus GmbH**, serving law firms through FastAPI, Twilio and Retell with real-time call handling and recovery paths.
 - **Multilingual delivery:** Extended German-language voice workflows to French, including locale-aware intake and mid-call language switching.
 - **Call recovery:** Designed queued attorney transfers that hold callers during connection retries and return them to the AI when a transfer cannot complete.
 - **AI platform:** Architected a multi-provider legal assistant with specialist agents, persistent conversations, tool execution and organisation-level data isolation; reduced median time to first streamed text by **70–75% in offline document-retrieval benchmarks**.
 - **Engineering quality:** Designed company-wide AI-assisted code-review infrastructure combining independent reviewers, specification checks, adversarial validation and incremental fix verification, with evaluation tooling for detection quality and cost.

SKILLS

System Architecture & Design

- **High-Level Design:** Cloud-native architecture, microservices, event-driven systems, scalability, and fault tolerance.
- **API & Data:** High-throughput API design (REST), real-time and batch data pipelines, hybrid data modeling (SQL/NoSQL).
- **Real-Time Systems:** Event-driven architectures (**Kafka**, **RabbitMQ**), streaming protocols (**SSE**, **WebSockets**) for low-latency LLM responses.

AI & Agentic Engineering

- **Agentic Frameworks:** **Lang-Graph** (multi-agent orchestration), **LangChain** (tool chains, memory), **Google ADK** (Agent Development Kit), **MCP** (Model Context Protocol), custom **MCP servers**, multi-provider gateway routing (**LiteLLM**).
- **Agentic Patterns:** **ReAct**, **Reflexion**, **Tree of Thoughts (ToT)**, Plan-and-Solve, Router/Worker, Self-Refine.
- **RAG & Retrieval:** Advanced chunking, semantic and hybrid search, query expansion, reranking, evaluation (**LLM-as-a-Judge**, **RAGAS**), **LlamaIndex**.
- **Vector Databases:** **ChromaDB**, **Pinecone**, **Weaviate**, **pgvector**, **Qdrant**, **FAISS**.
- **Memory Systems:** Semantic and episodic memory architectures, conversation compression, persistent memory patterns (**Mem0**).
- **LLM Operations:** Production integration (**Gemini**, **OpenAI**, **Anthropic**, **AWS Bedrock Converse API**), native tool calling, prompt caching (**cachePoint**), fine-tuning (**PEFT/LoRA**), structured outputs (**Instructor**), validation and guardrails (**Guardrails AI**).
- **Core ML:** Deep Learning (**PyTorch**), NLP (Transformers, Sentence-Transformers), Computer Vision (ViT, CNNs), Predictive Modeling (**Scikit-learn**, **XGBoost**, **LightGBM**), Time-Series Forecasting.

Voice AI & Real-Time Systems

Other selected consulting engagements

- **Founding AI Architect for a sales-automation startup**, designing a multi-agent system to automate customer communications across messaging platforms.
 - **Architecture:** Designed a **multi-agent microservices platform** using FastAPI, PostgreSQL, and Redis. Implemented a central **Router Agent** for intent classification that delegates tasks to specialized **Response Agents**, ensuring a modular and scalable design.
 - **Performance & Reliability:** Reduced response latency by approximately **98%** through Redis queue decoupling; resolved **80%** of inquiries autonomously with circuit breakers and fallback logic.
 - **Business Impact:** The core agentic logic of the MVP directly led to a **200% increase in sales conversion** for the initial beta clients.
- **Delivered a resume-parsing and job-matching platform** for a recruitment-technology client, automating candidate classification and semantic matching.
 - **AI & Backend:** Built a modular, microservices-based pipeline (FastAPI, Docker). Implemented a fine-tuned **Sentence-Transformer (PyTorch)** for classification and semantic matching agents using **LlamaIndex** and the **Google Gemini API**.
 - **Data & Frontend:** Designed a **hybrid database backend** — **PostgreSQL** for structured queries and **MongoDB** for unstructured data. Built the corresponding responsive frontend with **Next.js/React**.
 - **Performance:** Achieved a **93% F1 score** on role classification and reduced time-to-hire by **75%** for the client.
- **Architected and delivered a property-valuation platform** for a UK property-technology client, using **Google Cloud Platform (GCP)** for large-scale data processing.
 - **Architecture:** Designed a scalable **BigQuery** pipeline for high-volume joins and aggregations. Implemented asynchronous caching that reduced data-loading time by approximately **98%**, enabling faster experimentation.
 - **Modeling:** Systematically benchmarked five ML models (Linear Regression, XGBoost, LightGBM, RNN, Decision Trees) to identify the optimal balance of predictive accuracy and efficiency.
 - **Performance:** The final model placed **94% of predictions within 10%** of the actual sale price.
- **Delivered consulting for institutional financial-services clients**, building Python training material and R-based financial analysis tools that reduced manual reporting time by approximately **30%**.

Technical Lead & Mentor (Advanced Maths, Programming & AI/ML) Independent

📅 Sep 2014 – Present

📍 London, UK (Global Remote)

- **Provided technical mentoring globally** to students, researchers and professionals developing projects in AI/ML and data science.
- **Acted as a fractional Tech Lead for R&D projects**, including MSc dissertations and industry prototypes, driving architecture design while enforcing best practices through rigorous code reviews.
- **Outcomes:** Enabled mentees to secure admissions to Oxbridge/Russell Group/Ivy universities, win international Olympiads, and launch successful careers at leading tech companies (including FAANG).

- **Voice Agents:** **Retell AI** (multi-agent orchestration, language pinning, mid-call language switching), **OpenAI Realtime**, **Gemini Live**.
- **Telephony:** **Twilio** (Programmable Voice, TwiML, Queues, answering-machine detection, call forwarding), Aircall integration.
- **Real-Time Streaming:** WebSocket-based ASR/TTS pipelines and word-level timestamp extraction for audio-snippet playback.
- **Multilingual NLU:** German (Sie-form, umlauts, compound names, Cologne phonetic algorithm) and French (vous-form, accents, name particles, leading-zero phone normalisation) entity extraction across employment, family, traffic, criminal and tenancy law domains.
- **LLM Observability:** **Langfuse** (prompt registry, traces, A/B versioning, cost-spike incident response), **PostHog** (funnel events, language-switch tracking, latency telemetry), **Sentry** (logging integration, error capture).

Cloud, DevOps & Leadership

- **Leadership:** Technical leadership, mentoring, architectural design docs, stakeholder communication, cross-functional collaboration in regulated/enterprise environments (Tier-1 financial services).
- **Cloud Platforms:** **GCP** (BigQuery, Cloud Run, Vertex AI), **AWS** (Bedrock, Lambda, SageMaker, S3).
- **DevOps & Observability:** CI/CD (**GitHub Actions**), containerization (**Docker**, **Docker Compose**), tracing (**LangSmith**), observability (**Arize Phoenix**, **Weights & Biases**).
- **Tooling & Testing:** Git/GitHub, Linux Shell, **NGINX**, sandboxed execution (**E2B**); **Agent Testing** (unit tests for tools, integration tests for workflows, tracing with **LangSmith**, regression testing with golden datasets), **pytest**.
- **Frontend Prototyping:** **Next.js**, **React**, **TypeScript**, **Tailwind CSS**, **Streamlit**.

EARLIER EXPERIENCE

Lead Data Scientist (Indirect Contract)

Ministry of Health of Ukraine

Mar 2022 – Oct 2022 Kyiv, Ukraine (Remote)

- **Architected secure data pipelines** to process real-time signals from remote health monitors at a national scale.
- **Developed predictive ML models** (Scikit-learn, Python) to identify patients at high risk of diabetes complications, enabling earlier interventions and guiding public health strategy.
- **Generated key public health insights** by performing large-scale EDA on anonymized electronic health records (EHRs) to uncover novel correlations between biomarkers and disease progression.
- **Applied privacy controls** while processing sensitive health data and anonymised patient records.

Senior Machine Learning Engineer

Forecsys

Jan 2021 – Feb 2022 Remote

- **Led the design and production delivery of a real-time insurance fraud-detection system**, taking the product from concept through model-serving integration.
- **Architected high-throughput backend services and APIs** (Flask/FastAPI) to serve ML models with sub-second latency and high reliability under heavy load.
- **Championed new software engineering standards** (design patterns, automated testing), which reduced production bugs by ~30%.
- **Mentored junior data scientists** through implementation reviews and production deployment practices.

PROJECTS

Agentic Research Assistant for Scientific Literature

Personal R&D Project

Jan 2025 London, UK

- **Architected a multi-agent RAG system for scientific literature**, identifying reproducible experiments against laboratory equipment constraints.
 - **Orchestration:** Implemented a **LangGraph StateGraph** (Ingest → Triage → Extract → Synthesize → Rank) with conditional routing and retry on failures.
 - **Architecture:** Built a document ingestion pipeline (PyMuPDF, ChromaDB) and a hierarchical agentic workflow that performed a sequential triage, extraction, and synthesis process to generate a final feasibility report.
 - **Output:** Produced a feasibility report ranking candidate experiments with references to the supporting literature.

EDUCATION

Professional Certificate, Applied Machine Learning

The Machine Learning Institute (MLX)

Apr – May 2025

Intensive program covering Feature Fusion, Two-Tower Models, Transformers, Multimodality, PEFT/LoRA, RAG.

Bachelor's degree, Applied Mathematics and Physics

Moscow Institute of Physics and Technology (MIPT)

2013 – 2016

VOLUNTEER EXPERIENCE

Content Creator & Instructor - AQA GCSE Computer Science Course

Independent (YouTube & GitHub Project)

Sep 2024 – Present

Free online course (videos + GitHub repo) covering the AQA GCSE CS syllabus: algorithms, Python, systems, networks and supporting learning materials.

CERTIFICATIONS

Mathematics for Machine Learning Specialization

Coursera / Imperial College London

Mar 2021

LANGUAGES

English
Ukrainian
Russian

